

Unit 2 Notes

Transformations

How to use this file: Copy the notes carefully, then cover the worked examples and redo them without looking. For practice files, work first, then check the answer bank. Detailed solutions are included for Unit 3 through Unit 8 practice as requested.

Unit 2 Notes: Transformations

Big goal

A transformation is a function that moves points in the plane. The original figure is the **preimage**; the result is the **image**. Rigid motions preserve distance and angle measure, so they create congruent figures.

1. Transformations as functions

A transformation takes each input point (x, y) to exactly one output point (x', y') . You can write this with mapping notation, such as $(x, y) \mapsto (x + 3, y - 5)$. The arrow means “is sent to.” Prime notation names images: A maps to A' , B maps to B' , and so on.

2. Rigid motions and congruence

A **rigid motion** or **isometry** preserves lengths and angle measures. Translations, reflections, and rotations are rigid motions. A dilation is not a rigid motion unless the scale factor is 1 or -1 in a special orientation sense; in normal similarity work, dilation changes size and is not treated as congruence.

Two figures are congruent if a sequence of rigid motions maps one figure exactly onto the other. This definition connects transformations to triangle congruence and proof.

3. Coordinate rules for common transformations

Transformation	Rule	What changes
Translation by $\langle a, b \rangle$	$(x, y) \mapsto (x + a, y + b)$	Slides every point the same horizontal and vertical amount.
Reflection over x -axis	$(x, y) \mapsto (x, -y)$	Flips the sign of y .
Reflection over y -axis	$(x, y) \mapsto (-x, y)$	Flips the sign of x .
Reflection over $y = x$	$(x, y) \mapsto (y, x)$	Switches coordinates.
Reflection over $y = -x$	$(x, y) \mapsto (-y, -x)$	Switches coordinates and changes both signs.
Rotation 90° counterclockwise about origin	$(x, y) \mapsto (-y, x)$	Turns left one quarter-turn.
Rotation 180° about origin	$(x, y) \mapsto (-x, -y)$	Opposite point through origin.
Rotation 270° counterclockwise about origin	$(x, y) \mapsto (y, -x)$	Same as 90° clockwise.

4. Translations

A translation slides a figure. Every point moves along parallel paths by the same vector. Translations preserve lengths, angle measures, parallel lines, collinearity, and orientation. If $A(2, -1)$ is translated by $\langle -5, 4 \rangle$, then $A'(-3, 3)$.

5. Reflections

A reflection flips a figure over a line of reflection. The line of reflection is the perpendicular bisector of every segment connecting a point to its image. Reflections preserve size and shape but reverse orientation. For example, a clockwise order can become counterclockwise.

6. Rotations

A rotation turns a figure around a center by a given angle. Unless otherwise stated, positive rotations are counterclockwise. Rotations preserve size, shape, angle measure, parallelism, and orientation.

7. Sequences of transformations

A sequence is a composition. Order matters. Reflecting then translating can give a different result from translating then reflecting. To describe a sequence, give enough detail: type of transformation, line of reflection or center and angle, translation vector, and order.

Example: Sequence

Start with $P(3, 4)$. Reflect over the y -axis: $(3, 4) \mapsto (-3, 4)$. Then translate by $\langle 2, -1 \rangle$: $(-3, 4) \mapsto (-1, 3)$. The final image is $(-1, 3)$.

8. Symmetry

A figure has symmetry if a nontrivial rigid motion maps the figure onto itself.

- **Line symmetry:** A reflection maps the figure onto itself.
- **Rotational symmetry:** A rotation less than 360° maps the figure onto itself.
- **Order of rotational symmetry:** The number of times the figure maps onto itself in one full turn.

A square has four lines of reflection symmetry and rotational symmetry of order 4, with smallest positive rotation 90° . A non-square rectangle has two lines of reflection symmetry and rotational symmetry of order 2.

9. Parallel lines and transversals

A transversal is a line that intersects two or more lines. When a transversal crosses parallel lines, angle relationships appear because translations slide one intersection to the other.

Angle pair	Relationship when the lines are parallel
Corresponding angles	Congruent. Same relative position at each intersection.
Alternate interior angles	Congruent. Inside the two lines, opposite sides of transversal.
Alternate exterior angles	Congruent. Outside the two lines, opposite sides of transversal.
Same-side interior angles	Supplementary. Inside the two lines, same side of transversal.
Same-side exterior angles	Supplementary. Outside the two lines, same side of transversal.
Vertical angles	Congruent, whether or not the lines are parallel.
Linear pairs	Supplementary, whether or not the lines are parallel.

10. Proving lines are parallel

The converses are just as important. If a pair of corresponding angles is congruent, the lines are parallel. If a pair of alternate interior angles is congruent, the lines are parallel. If same-side interior angles are supplementary, the lines are parallel.

Do not assume parallel lines from a picture

Parallel marks, written information, or a theorem must justify parallel lines. A drawing can be misleading.

11. Worked examples

Example 1: Transformation rule

Apply $(x, y) \mapsto (-y, x)$ to $A(-2, 5)$. Since this is a 90° counterclockwise rotation, $A'(-5, -2)$.

Example 2: Transversal algebra

If alternate interior angles measure $5x + 7$ and $3x + 35$ on parallel lines, set them equal:

$$5x + 7 = 3x + 35 \Rightarrow 2x = 28 \Rightarrow x = 14.$$

Each angle is 77° .

Example 3: Proving congruence by motion

If triangle ABC can be reflected over a line and then translated to land exactly on triangle DEF , then the triangles are congruent, because reflections and translations are rigid motions.

Mastery checklist

You are ready for Unit 2 when you can apply coordinate rules quickly, describe transformation sequences, identify symmetry, use angle relationships formed by parallel lines and transversals, and prove lines parallel using converse angle theorems.